LESSON PLAN · SCRATCH (GR 6–8) ADVANCED: ALGORITHMS LAB

Lesson 2: Binary Search: Divide and Conquer

Module 1 | 45 minutes | Binary search | Grades 6–8 · ages 12–14

Objective: Implement binary search on a sorted list using low, high and mid, and compare its efficiency with linear search.

Success criteria

- I can implement binary search.
- I can trace low, high and mid.
- I can compare its efficiency with linear search.

Key vocabulary

- **binary search**: Repeatedly halving a sorted list to find a target.
- **divide and conquer**: Solving a problem by splitting it.
- **mid**: The middle position of the current range.
- **logarithmic**: Grows very slowly as input grows (halving).

You will need

- Projector with Kitabash (lesson video, Code Lab)
- Devices: 1 per learner or pair
- Printed worksheet (optional)

Timed lesson plan (45 min)

Time	Stage	What happens
5 min	**Warm-up (unplugged)**	Phone-book race (5 min): sorted cards 1–32; one team linear, one binary; count flips.
5 min	**Watch**	Play the lesson video with captions; pause at each heading and ask a question.
8 min	**Build along**	Binary search 1–100: learners follow the steps.
14 min	**Perform**	Code Lab: binary search, Halving power, Trace binary search
9 min	**Probe**	activity
4 min	**Wrap-up**	Exit ticket: Write the formula for mid.

Unplugged warm-up

Phone-book race (5 min): sorted cards 1–32; one team linear, one binary; count flips.

Teaching the idea

Binary search on sorted arrays maintains [low, high]; mid compared with target; range halves each iteration → O(log n). Correct loop condition (low ≤ high) and ±1 updates avoid infinite loops/misses.

Analogy: Guess my number: always guess the middle of what's left.

Live demo: Play "guess my number 1–100" with the class using halving; count guesses.

Questions to ask

Ask	Listen for
Why must the list be sorted?	To know which half to discard.
What happens if you forget +1/−1?	Infinite loop or misses.
Max steps for 1–100?	7.

Differentiation

Support
Provide a trace table and the mid formula.

Stretch
Run both searches for 20 random targets and average the steps.

Common misconceptions

Misconception	How to fix it
Binary search works on any list	Only sorted lists.
mid = high / 2	mid = (low + high) / 2.

Assessment

Use the Code Lab results (auto-checked), the quick checks and the Probe answers. Written answers arrive in your review queue.

Exit ticket
Write the formula for mid.

Homework
Play "guess my number" 1–1000 with someone and count your guesses using halving.

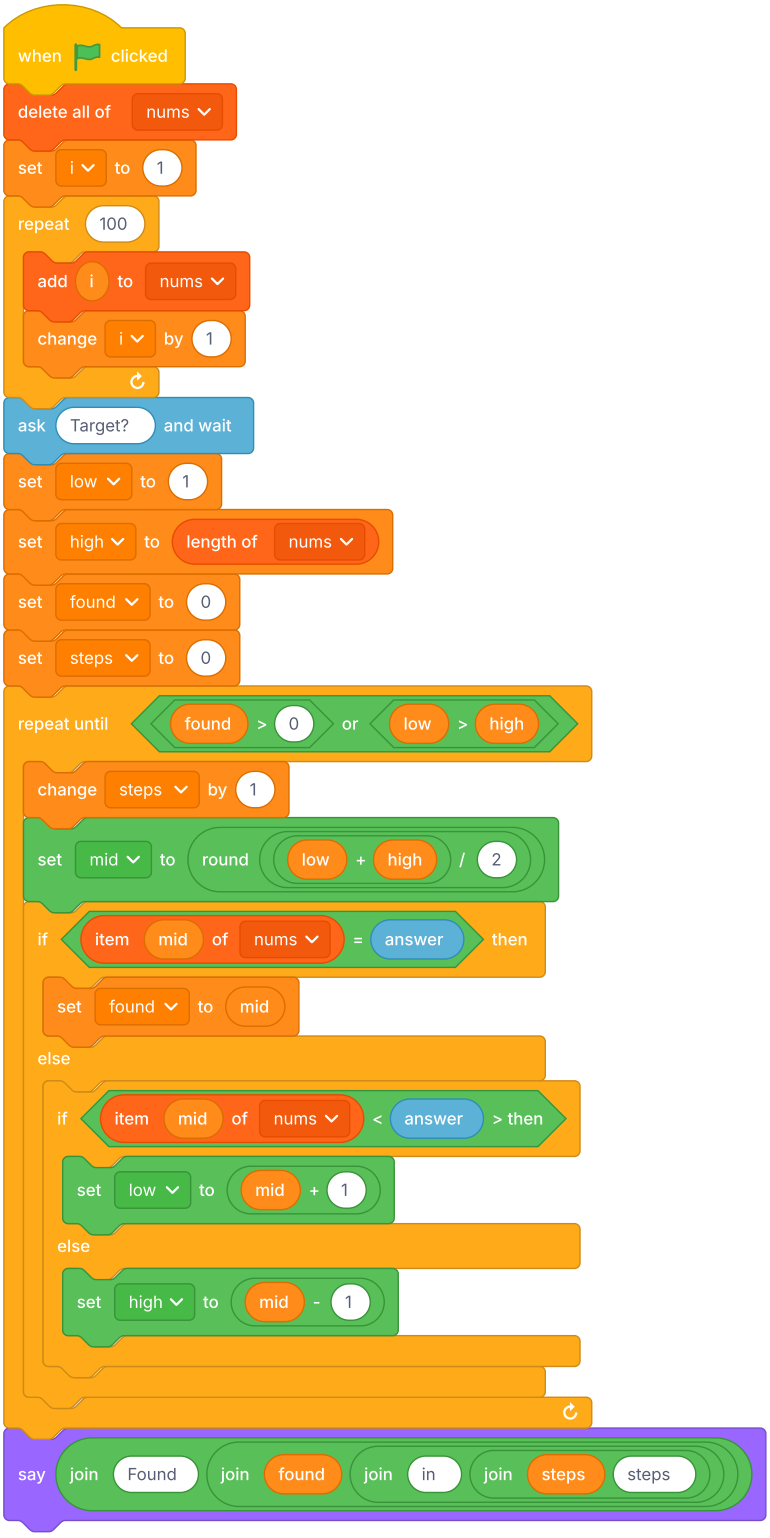
Answer key — Kitabash activities

1. Code Lab: binary search (perform · code lab)

Goals checked: Finds 73 · In at most 7 steps · Finds the edge case 100 · Uses mid

One solution:

Seeker



2. Halving power (perform · multiple choice)

Answer: About 20

3. Trace binary search (perform · fill blank)

Answers: 49 · 26 · sorted

4. Explain your thinking (probe · free response)

Look for: sorted, steps, binary. **Example:** Binary search takes far fewer steps — about 7 for 100 items — but the list must be sorted. Linear search works on any list but may take as many steps as there are items. I would use binary search for a sorted phone book and linear search for a short, unsorted shopping list.

Answer key — worksheet

1. **Key words:** binary search = Repeatedly halving a sorted list to find a target. · divide and conquer = Solving a problem by splitting it. · mid = The middle position of the current range. · logarithmic = Grows very slowly as input grows (halving).
2. **Plan your code:** Goals: Finds 73; In at most 7 steps; Finds the edge case 100; Uses mid.
3. **Halving power:** A. About 20 — 2^20 ≈ 1,000,000, so about 20 halvings.
4. **Trace binary search:** 49 · 26 · sorted
5. **Explain your thinking:** Binary search takes far fewer steps — about 7 for 100 items — but the list must be sorted. Linear search works on any list but may take as many steps as there are items. I would use binary search for a sorted phone book and linear search for a short, unsorted shopping list.

Kitabash original curriculum. Scratch is a project of the Scratch Foundation in collaboration with the MIT Media Lab; this course is not affiliated with or endorsed by them.